

1 Objective

To evaluate precision and recall of hybrid search application using BioPortal as the ground-truth reference.

Relevant Entities	Retrieved Entities
Platform: BioPortal Search term: "Consecutive hemorrhage" http://purl.obolibrary.org/obo/MP_0001914 http://purl.obolibrary.org/obo/MONDO_0013792 http://purl.obolibrary.org/obo/MONDO_0005099 http://purl.obolibrary.org/obo/MP_0006052 http://purl.obolibrary.org/obo/MP_0006051	Platform: Hybrid Search Search term: "Consecutive hemorrhage" http://purl.obolibrary.org/obo/MONDO_0005099 http://purl.obolibrary.org/obo/MP_0006052 http://purl.obolibrary.org/obo/MP_0006051 http://purl.obolibrary.org/obo/MP_0005245 http://purl.obolibrary.org/obo/MP_0003888

Figure 1: The search results in BioPortal and Hybrid Search platforms are shown. The common entities found in both platforms are highlighted by the grey box.

2 Approach

BioPortal APIs enable searching different types of resources. We develop the reference datasets (ground-truth) as follows.

- Identify target ontologies. E.g. MONDO, CHIRO, etc.
- Using BioPortal API, search for terms in the target ontologies.
- Extract the relevant fields and save the output (**golden** dataset).

Subsequently, the same queries used for curating the datasets from BioPortal are run in the Hybrid Search platform for comparison as follows.

- Using hybrid search APIs, run the same queries for finding entities in the target ontologies.
- Compare the results with the golden dataset.

3 Datasets

Descriptions of the datasets curated using BioPortal API are provided in Table 1. The number of items retrieved from BioPortal for each query was

capped at $k = 20$. The value of k was arbitrarily picked, and can be changed as needed in the future.

Table 1: Dataset descriptions.

Dataset ID: DS01
Filename: bp_class_search_chiro_mondo_at_least_n_minus_one.jsonl
Description: For a multi-word search query containing N_{terms} unique terms, at least $N_{terms} - 1$ of those terms are found in the fields ‘definition’, ‘prefLabel’, and ‘synonym’ (accumulative, not necessarily in each field).

4 Results

The performance evaluation of hybrid search platform started in different retrieval and reranker modes is summarized in Table 2. As described in Section 3, the value of k represents the maximum number of items curated for each query in the reference (BioPortal) dataset. The same value of k was used as *max_results* parameter when calling the hybrid search endpoints. Let $L_{dataset}$ denote the number of items in the dataset, Q05 and Q95 denote the 5-th and 95-th quantiles.

Table 2: Test results with $L_{dataset} = 30$ and $k = 20$.

Dataset ID	Retrieval mode	Reranker type	Results
DS01	Hybrid	ensemble	Mean precision@k=0.233 Q05 precision@k=0.0 Q95 precision@k=1.0 Mean recall@k=0.109 Q05 recall@k=0.0 Q95 recall@k=0.7
DS01	BM25-only	biomedical	Mean precision@k=0.217 Q05 precision@k=0.0 Q95 precision@k=1.0 Mean recall@k=0.101 Q05 recall@k=0.0 Q95 recall@k=0.7